Making the Most of Little Time: Probabilistic Marine Heatwave Detection

THE PROBLEM: The 30-Year Barrier

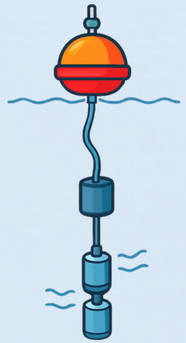


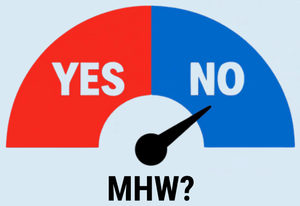
The 30-Year Standard

Conventional marine heatwave (MHW) detection requires long-term data for reliable climatological baselines

Subsurface Data Gaps

Many in-situ subsurface ocean records are too short or fragmented for traditional baseline calculations. However, understanding subsurface MHWs is important.



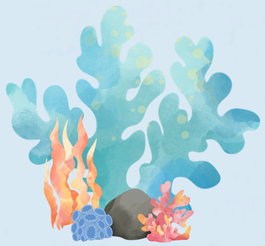


The Binary Limitation

Conventional “yes/no” detection fails to account for uncertainty for short time series.

Ecological Significance

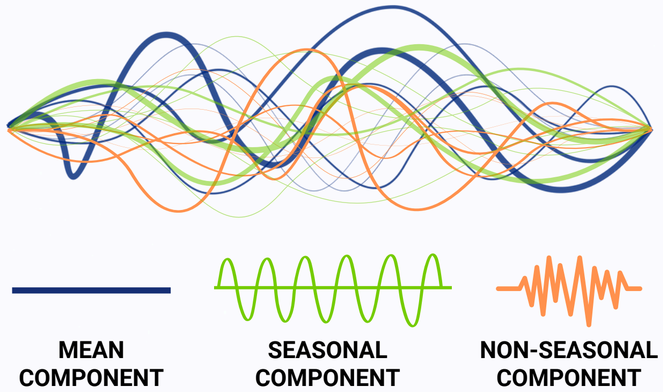
Inappropriate baselines can overlook thermal stress, distorting our understanding of ecosystem impacts.



THE SOLUTION: The Bayesian Probabilistic Model

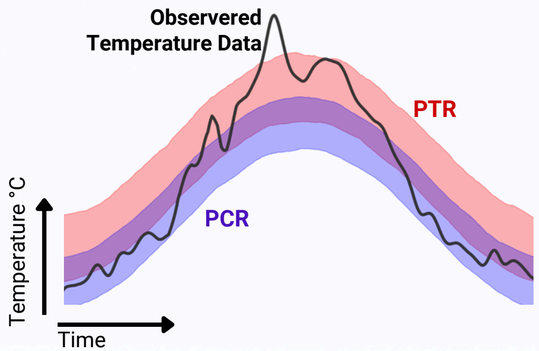
PyMC Signal Decomposition

Decomposes temperature signal into:

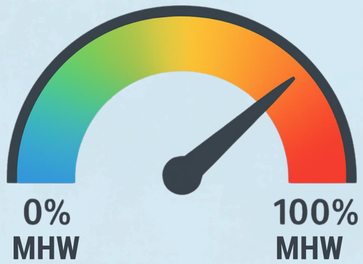


Probabilistic Baselines

Using only the mean and seasonal component, the framework learns a distribution of plausible climatologies (PCR, Probabilistic Climatology Range) and thresholds (PTR, Probabilistic Threshold Range). Example based on 7yr subset:



THE RESULT: Reliable Detection with Short Term Time Series

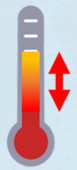


The Heatwave Probability Score (HPS)

Replaces “yes/no” labels with a 0–100% Heatwave Probability Score by quantifying observed temperature against the PTR.

Capturing Uncertainty

Provides probabilistic ranges for intensity and duration, guiding ecological impact analysis.



Comparison MHW Detections for 2010/11 MHW event at South WA site: 30yr Baseline vs. 3yr subset

Metric	30yr Baseline Hobday-Method	3yr Subset Probabilistic Method
Total Duration	147 Days within 5 events	139 Days within 5 events
Maximum Intensity	3.8°C	2.3 – 3.7°C
Cumulative Intensity	255.2°C days	50–249°C days